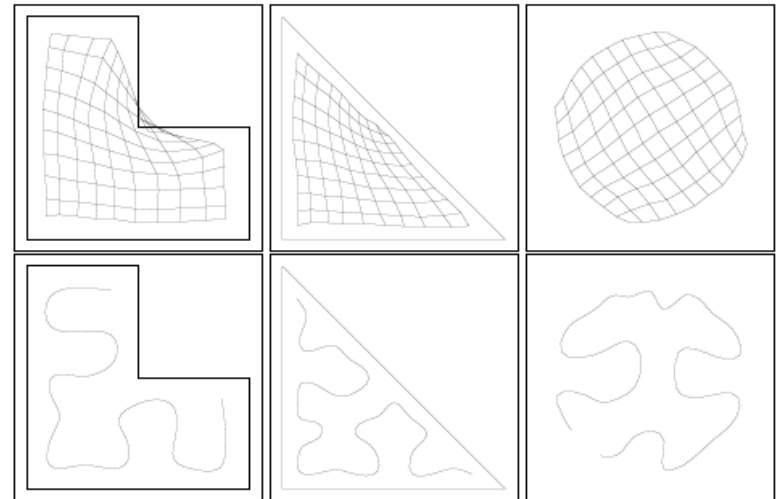


Neural Networks and Machine Learning

Unsupervised Learning and Self-Organising Maps

- Introduction to unsupervised learning
- Competitive learning
- Feature maps / SOMs
- Topographic maps in the brain
- Von der Malsburg's Model
- Kohonen Algorithm
- Applications of the Kohonen Algorithm
- Elastic Net Algorithm



Unsupervised learning

- Supervised learning assumes the target output is known.
- In many cases, this is not biologically realistic.
- Unsupervised learning is learning without a teacher.
- No feedback what the output should be, or even if it is correct.
- Can discover patterns, features, regularities, categories in the environment.
- Requires redundancy in the input data.
- Important applications include the formation of **topographic maps** and the **clustering** of inputs by **competitive learning**.

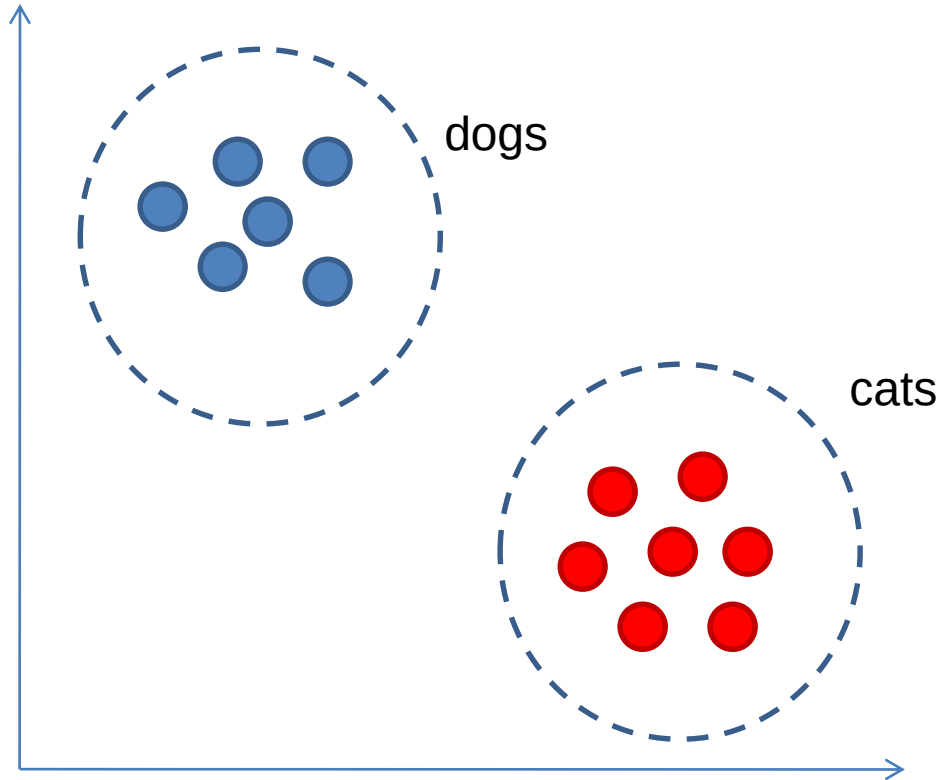


Cats and dogs can be represented by vectors



001100001
=
001110000
110001111 . . .

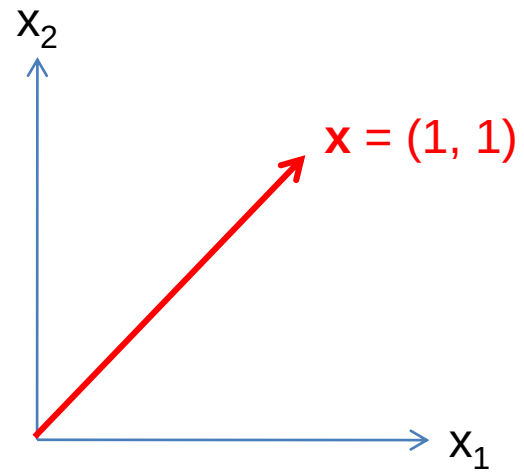
Clusters in input space



Want one output unit that responds preferentially (or only) to dogs and another output unit that responds preferentially to cats.

How can we determine the weight vectors to these output units?

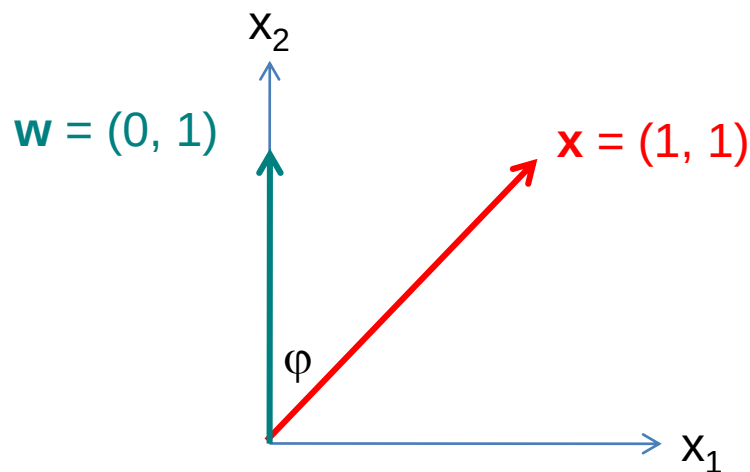
Vectors and inner products



Vectors and inner products

Activation of an output unit is given by the weighted sum of inputs $\sum w_i x_i$

The weighted sum of inputs can also be represented using the inner product (or dot product) of the input vector and the weight vector: $\sum w_i x_i = \mathbf{w} \cdot \mathbf{x}$



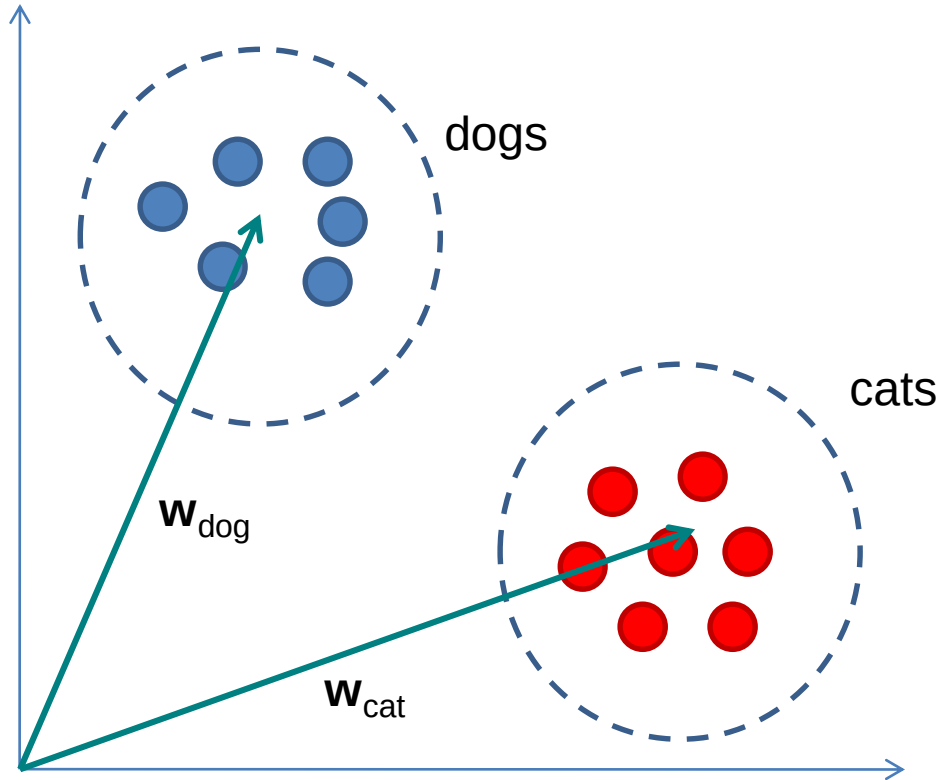
$$\mathbf{w} \cdot \mathbf{x} = \|\mathbf{w}\| \|\mathbf{x}\| \cos \varphi$$

$$\mathbf{w} \cdot \mathbf{x} = 0 \times 1 + 1 \times 1 = 1$$

The inner product increases with increasing similarity between the two vectors .

This means that an output unit is maximally activated by inputs that are similar to its weight vector.

Specialising on clusters in input space

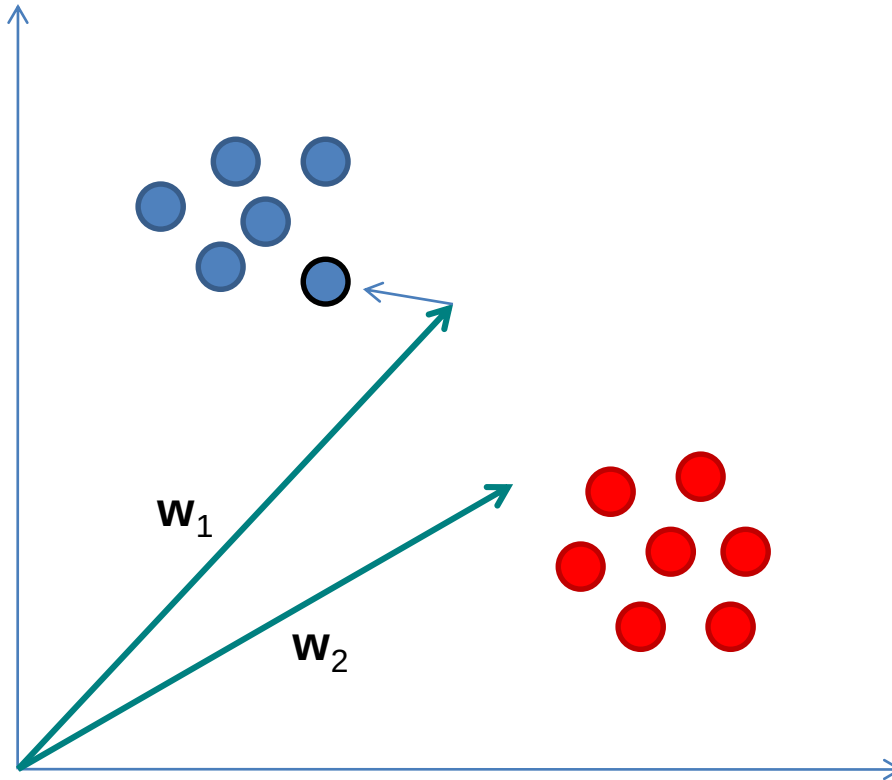


The weight vector of the “dog” unit should point to the centre of the “dog” cluster, and the weight vector of the “cat” unit should point to the centre of the “cat” cluster.

How can we learn the required weight vectors?

Moving weight vectors in input space

How can we learn the required weight vectors?



Start with random initial weight vectors.

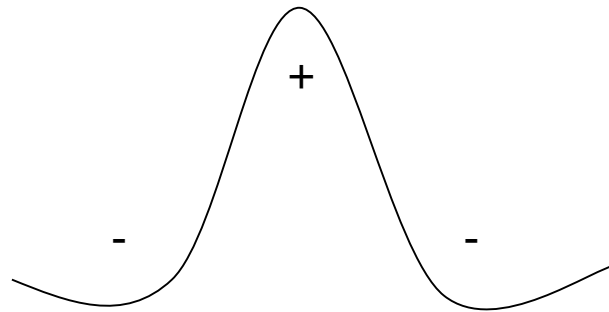
Present an input pattern and determine the winning output unit with the strongest activation.

Move the weight vector of the winning unit towards the input vector.

Determining the winning output unit

The winning output unit can be determined in several different ways.

Most biologically realistic, but least efficient way of choosing the winner: add long-range (ideally global) lateral inhibition and short-range (or self) excitation to the network. A similar (but more local) connectivity pattern is often found in the brain and described by a Mexican hat function.



A more efficient way of finding the winner

More efficient: calculate the activation of the output units as a function of the inputs x_j and weights w_{ij} and hand-pick the output unit with the largest activation y_i :

$$\text{Output activation: } y_i = x_1 w_{i1} + x_2 w_{i2} \dots = \sum_j x_j w_{ij} = \vec{x} \cdot \vec{w}_i$$

$$\text{Winner: } y_{i^*} = \max(y_i)$$

$$\text{Weight update only for the winner: } \vec{w}_{i^*} \rightarrow \vec{w}_{i^*} + \alpha(\vec{x} - \vec{w}_{i^*})$$

where α is the learning rate. This moves the weight vector $\vec{w}_{i^*}$ of the winning output unit towards the current input vector $\vec{x}$.

This mechanism requires normalised weight vectors and benefits from normalised input vectors. The vectors can be normalised by reinforcing that they all have the same length, e.g.:

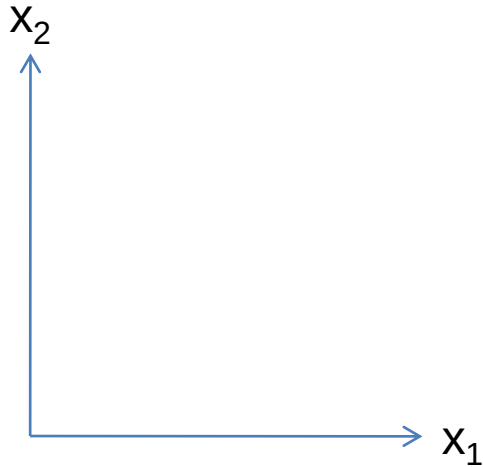
$$\sum_j x_j^2 = 1$$
$$\sum_j w_{ij}^2 = 1$$

A more efficient way of finding the winner

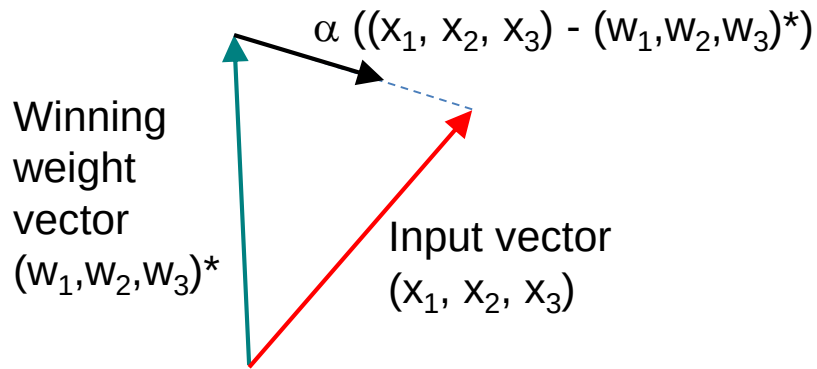
$$y_i = x_1 w_{i1} + x_2 w_{i2} \dots = \sum_j x_j w_{ij} = \vec{x} \vec{w}_i$$

$$y_{i^*} = \max(y_i)$$

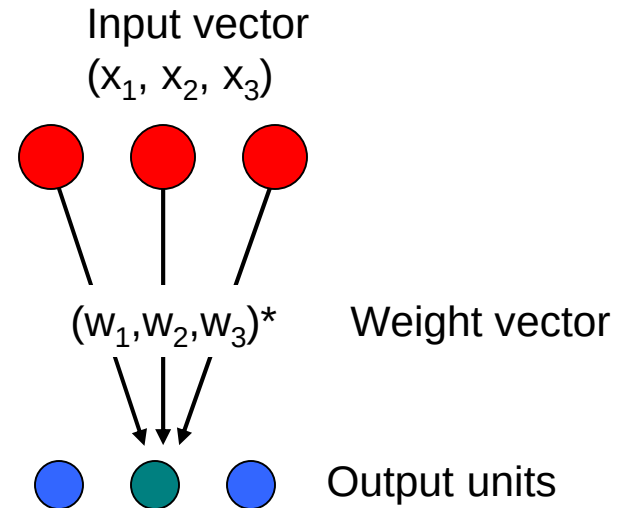
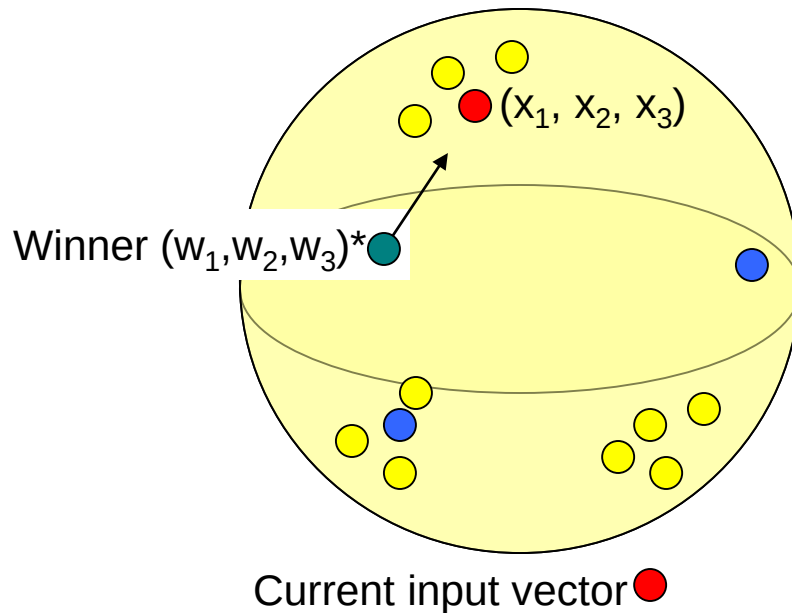
$$\vec{w}_{i^*} \rightarrow \vec{w}_{i^*} + \alpha(\vec{x} - \vec{w}_{i^*})$$



Competitive learning moves weight vectors in input space



Input space



Normalised vectors move around on a (hyper-) sphere.

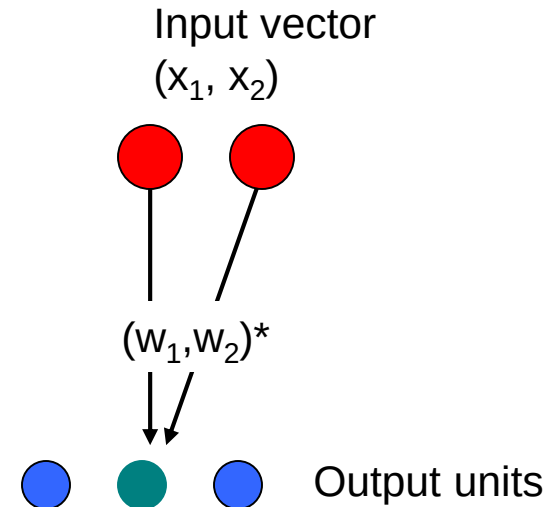
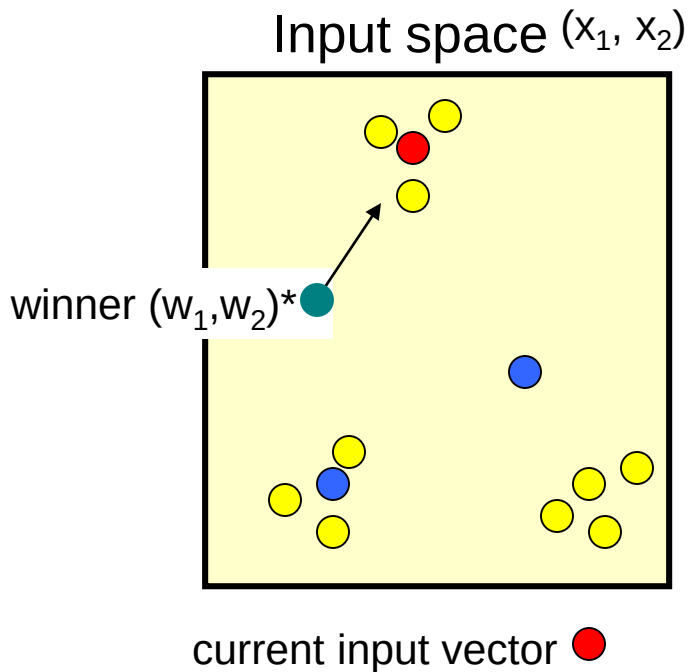
$$\vec{w}_{i^*} \rightarrow \vec{w}_{i^*} + \alpha(\vec{x} - \vec{w}_{i^*})$$

The most efficient way of finding the winner

Instead of calculating output activations we can simply compare the weight vectors of the output units with the current input vector and choose the closest one.

For the winning output (with weight vector $\mathbf{w}_{i*}$): $\|\vec{X} - \vec{w}_{i*}\| \leq \|\vec{X} - \vec{w}_i\|$

This abolishes the need for normalization and allows more freedom of movement.
For 2D inputs and weights:



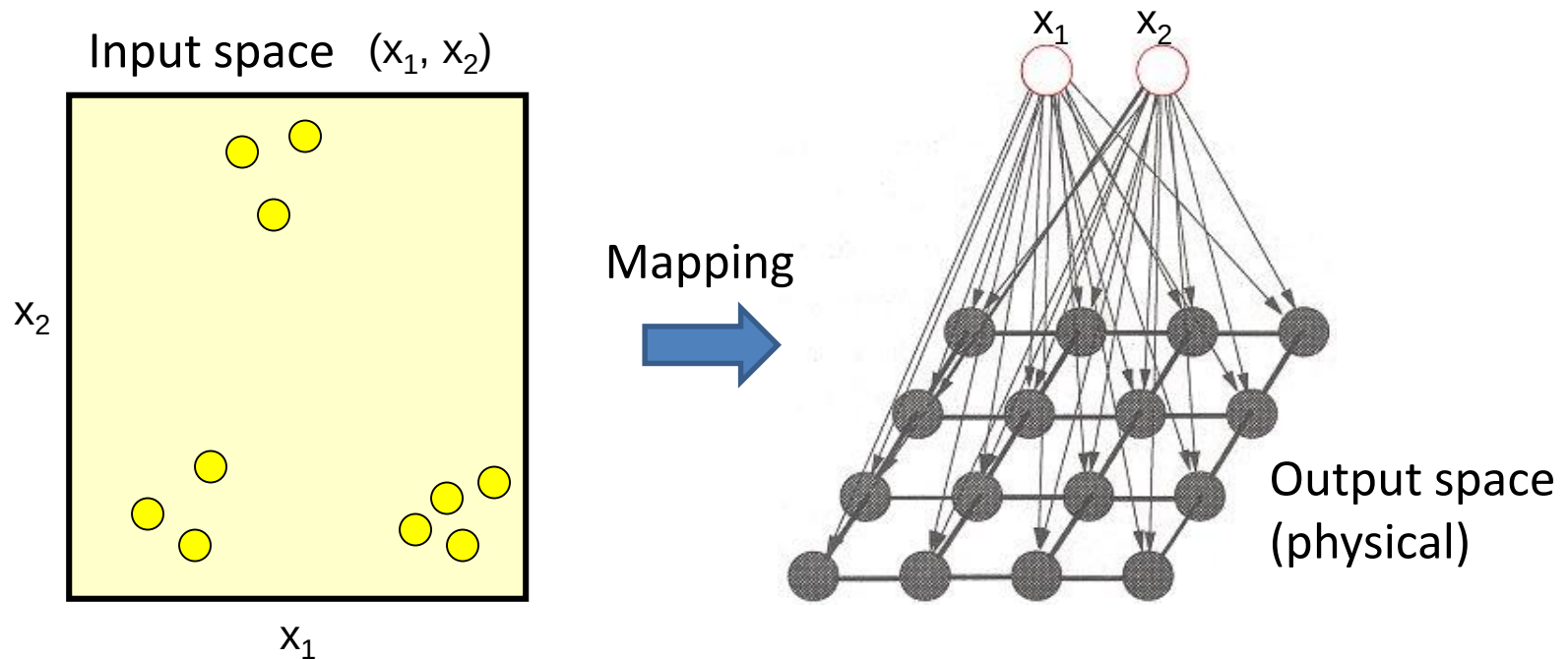
How would this look for normalised 2D vectors?

Feature maps

Feature maps (often called self-organising maps, SOMs) involve a special form of competitive learning where the output units are arranged in an ordered way in physical space.

Usually the output units form a regular 2D grid, but other arrangements are possible.

The aim is for output units that are close to each other in physical space to respond to inputs that are similar (i.e. close together in input space).



This resembles the topographic maps that are found in the brain.

Topographic maps in the brain

In a topographic map, similar inputs activate neighbouring outputs.

In the simplest case, the topographic map is neighbourhood preserving so that neighbouring inputs are mapped onto neighbouring outputs.

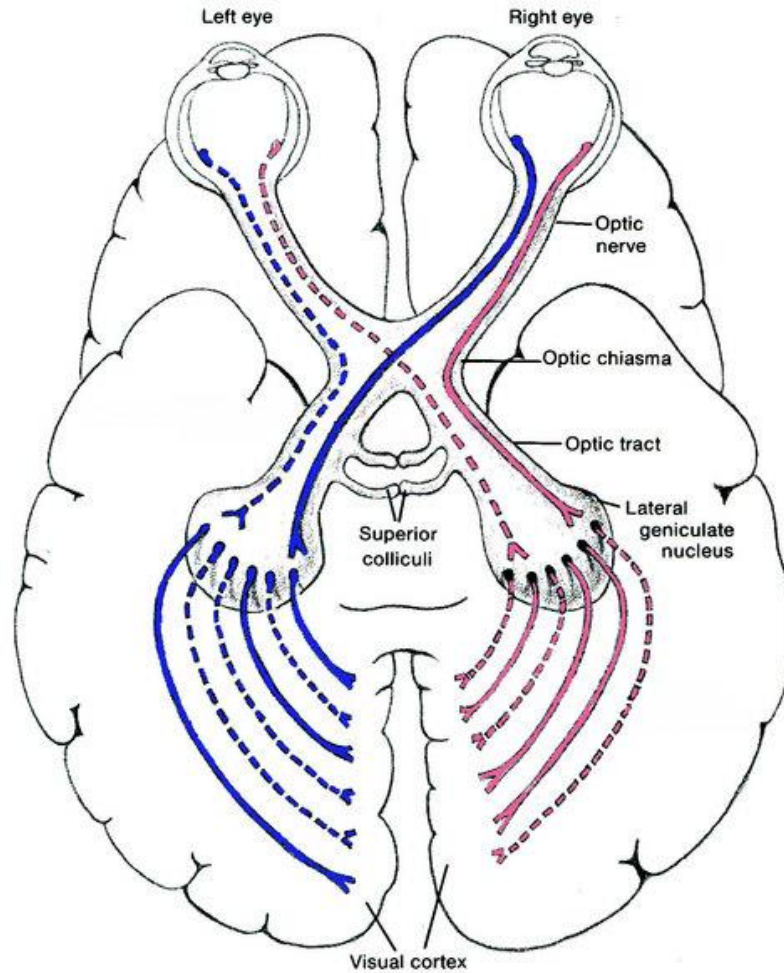
Examples for topographic maps in the brain:

- Retinotopic map in the visual cortex
- Orientation map in the visual cortex
- Somatotopic maps in the sensory and motor cortex
- Tonotopic map in the auditory cortex

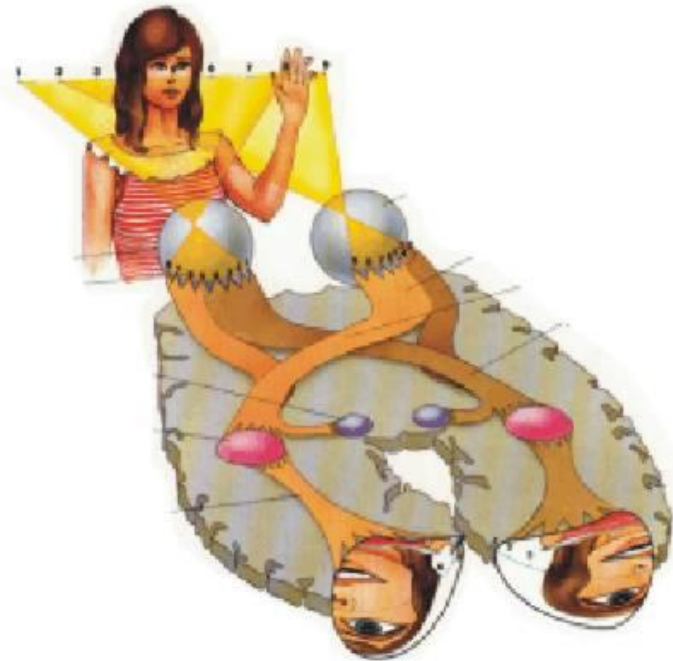
Advantages of topographic maps:

- Noise resistance
- Minimization of wiring costs

The retinotopic map in the visual cortex (1)



visual field

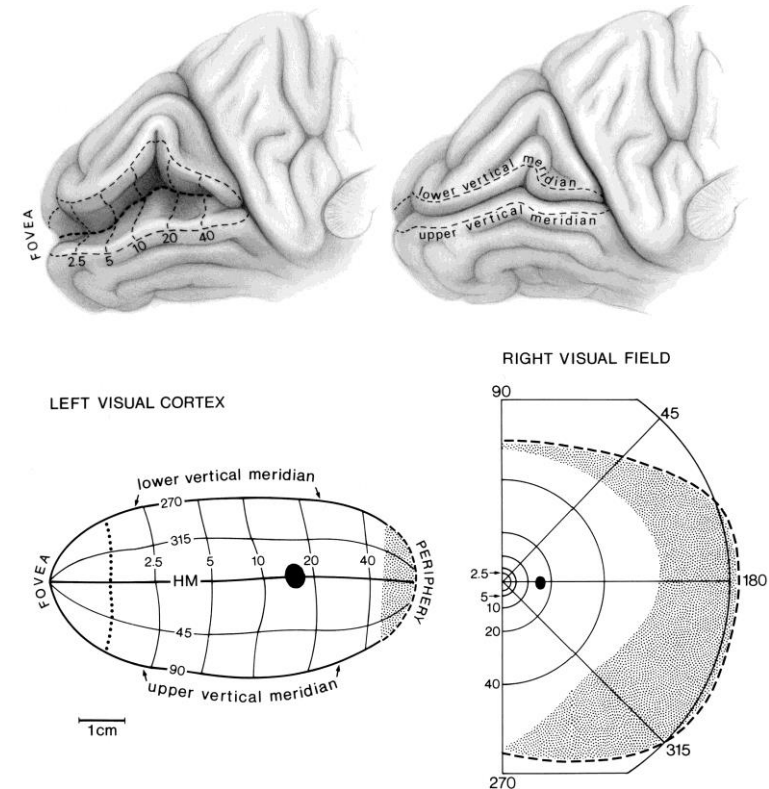
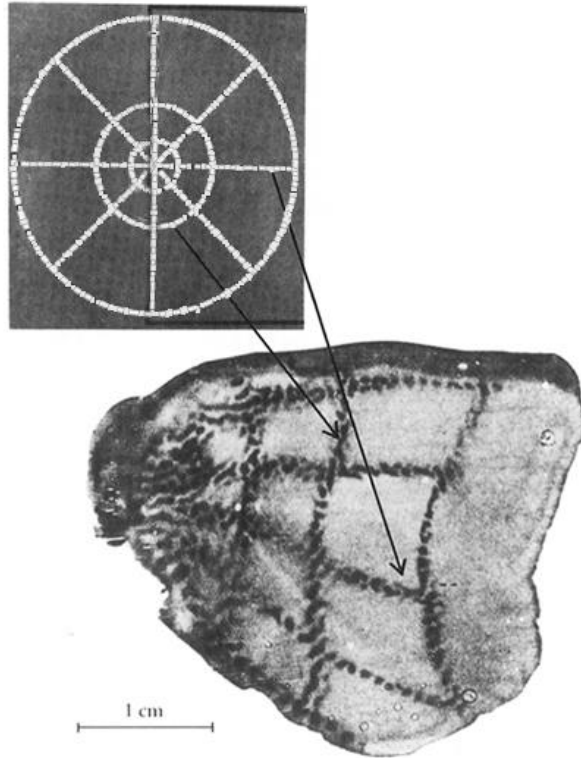


visual cortex

The retinotopic map in the visual cortex (2)

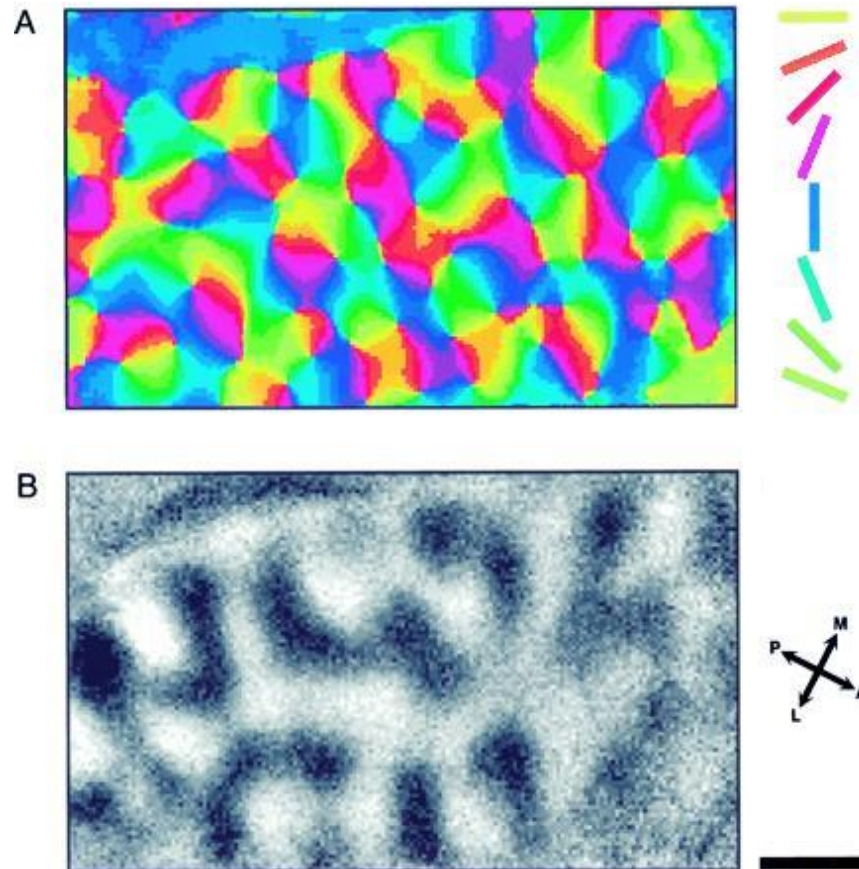
Adapted From *Eye, Brain and Vision*,
by D.H. Hubel, 1988, Sci Am Library.

In this experiment by Roger Tootell, the target-shaped stimulus with radial lines was centered on an anesthetized macaque monkey's right visual field for 45 minutes after injection with radioactive 2-deoxyglucose. One eye was held closed. The lower picture shows the labeling in the striate cortex of the left hemisphere. This autoradiograph shows a section parallel to the surface; the cortex was flattened and frozen before sectioning. The roughly vertical lines of label represent the (semi)circular stimulus lines; the horizontal lines of label represent the radial lines in the right visual field. The hatching within each line of label is caused by only one eye having been stimulated and represents ocular-dominance columns.



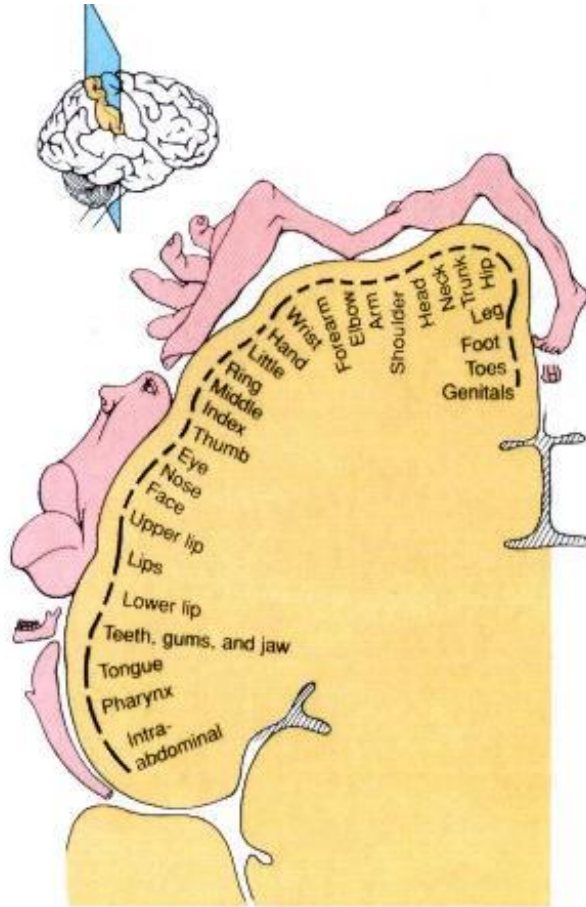
- Neighbouring areas in the visual field are processed by neighbouring areas of cortex.
- A large area of cortex is dedicated to the area surrounding the fovea (centre of gaze, contains highest density of receptors).

The orientation map in the visual cortex

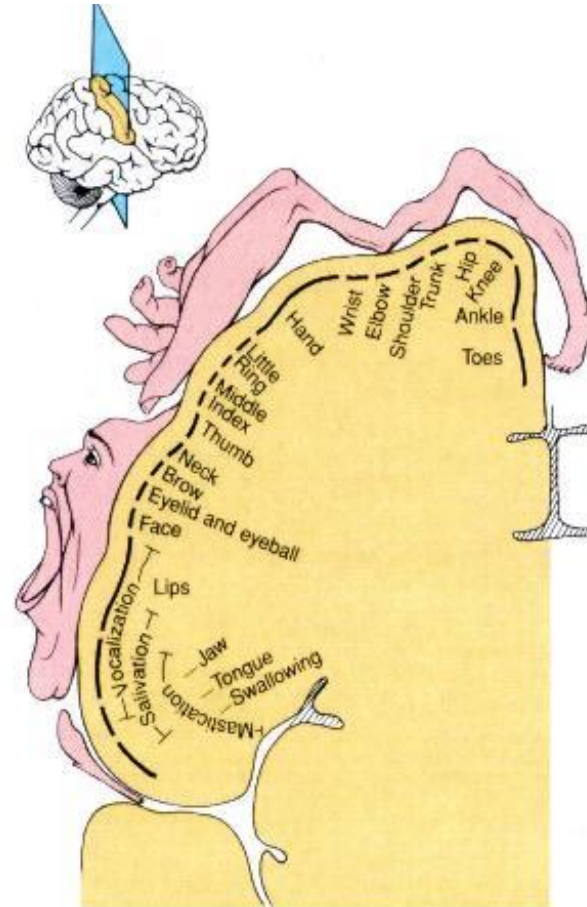


- (A) Areas in the visual cortex respond preferentially to bars with different orientations. The orientation preference varies gradually around discontinuities called pinwheels.
- (B) Ocular dominance columns: alternating areas of cortex respond to input from the left or right eye.

Somatotopic maps in sensory and motor cortex



(a) Somatosensory cortex in right cerebral hemisphere

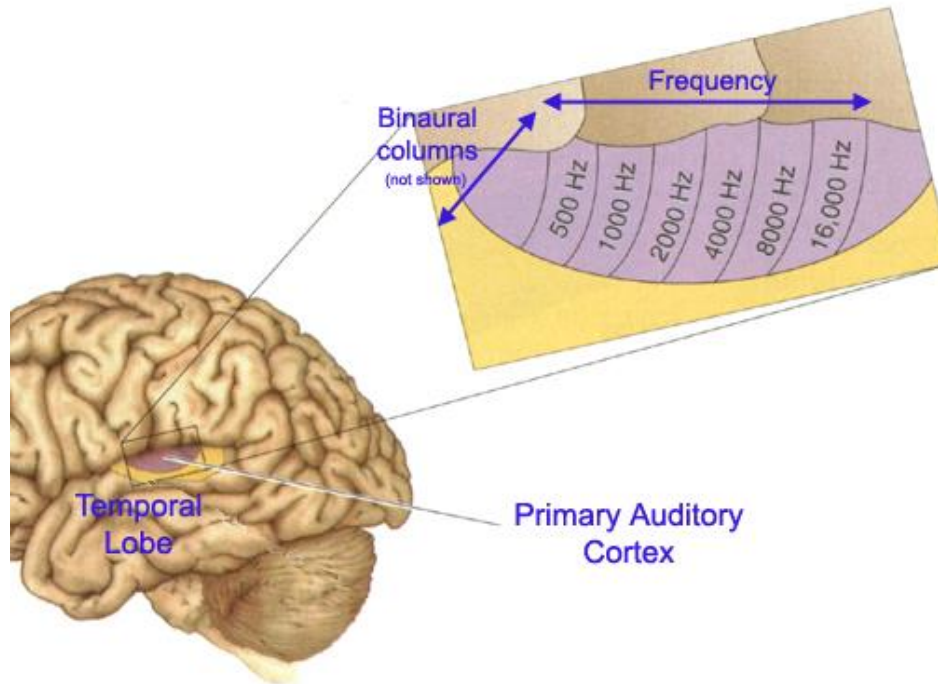


(b) Motor cortex in right cerebral hemisphere

Distorted “homunculus”: skin areas with higher densities of receptors (e.g. lips, fingers) are represented by larger areas of cortex.

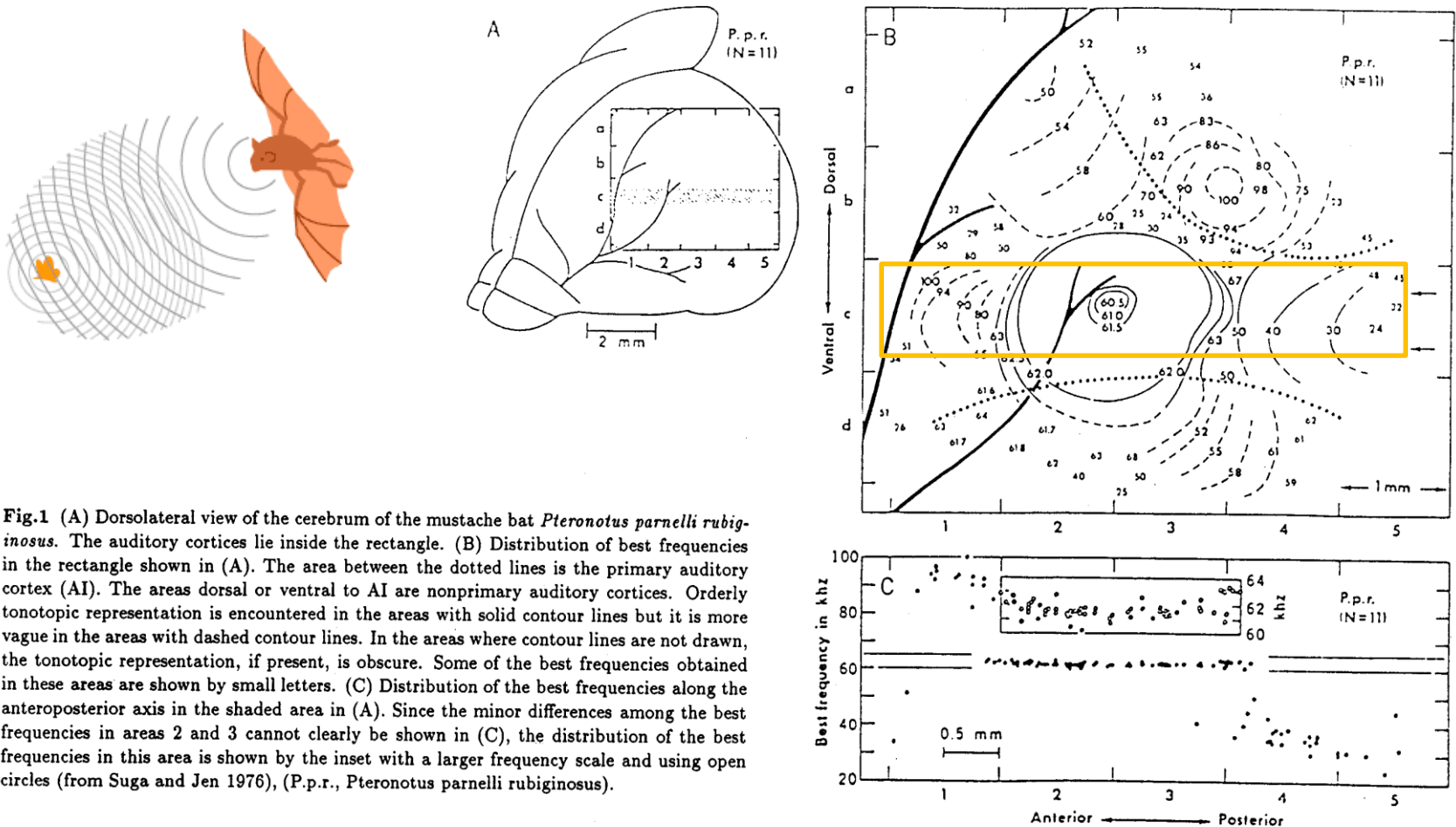
Tonotopic map in the auditory cortex

Tonotopic Map Has Columnar Organization



The auditory cortex contains a continuous and monotonic representation of sound frequencies.

The tonotopic map in bats



Mustached bats use sonar signals with frequencies around 61kHz for echolocation. These frequencies are over-represented in the auditory cortex.

Modelling the formation of an orientation map

Von der Malsburg (1973): cortical layer with 169 units receives input from retina with 19 units. Output units in the cortical layer excite their nearest neighbours and inhibit (mediated by a separate set of units) a slightly larger local region.

Activate retina with binary patterns that represent light bars. Calculate the resulting cortical activity and update the weights vectors by Hebbian learning so that they become more similar to the input vectors.

Inputs to the retina

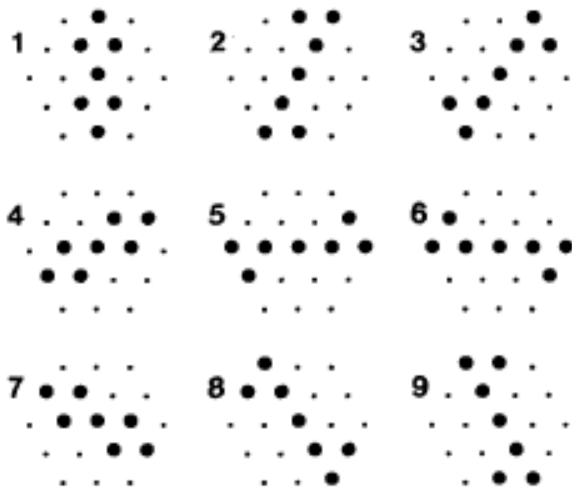


Fig. 5. The standard set of stimuli used on the model "retina". Large and small dots represent active and non-active fibres respectively

Resulting orientation map showing the preferred orientation for every cortical unit

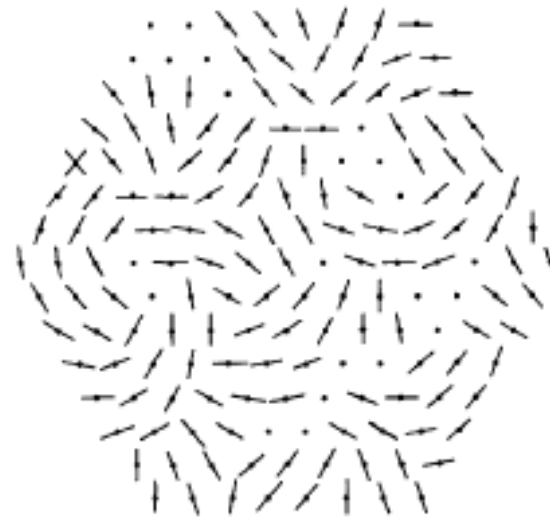


Fig. 13. View onto the cortex after 100 steps of learning

The Kohonen Algorithm (1982)

Engineering approach to map formation: less biologically realistic, more efficient. Used to map an N-dimensional input space onto a set of output units that is arranged in a physical 1D or (more commonly) 2D array. Aim: make the weight vectors of the output units similar to the input vectors so that similar input vectors activate neighbouring output units.

The algorithm:

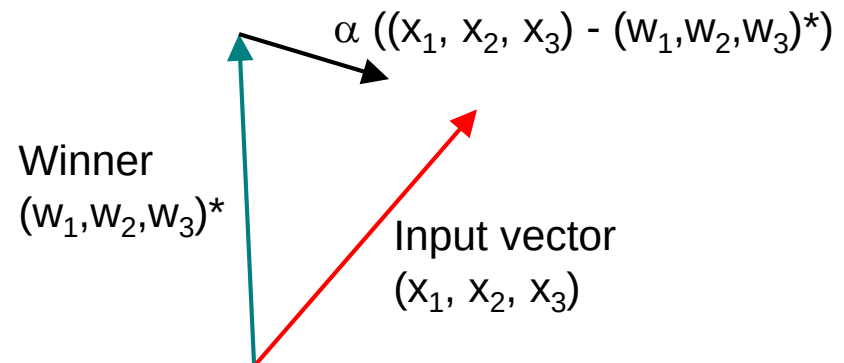
1. Start with small random weights.
2. Present a randomly chosen input vector $\mathbf{x}$.
3. Find the output unit whose weight vector $\mathbf{w}_{i^*}$ is closest to the input vector $\mathbf{x}$:

$$\|\vec{x} - \vec{w}_{i^*}\| \leq \|\vec{x} - \vec{w}_i\|$$

4. Update the weight vector of the winning output unit **and its neighbours** so that they become more similar to the input vector (with learning rate α):

$$\vec{w}_i \rightarrow \vec{w}_i + \alpha (\vec{x} - \vec{w}_i)$$

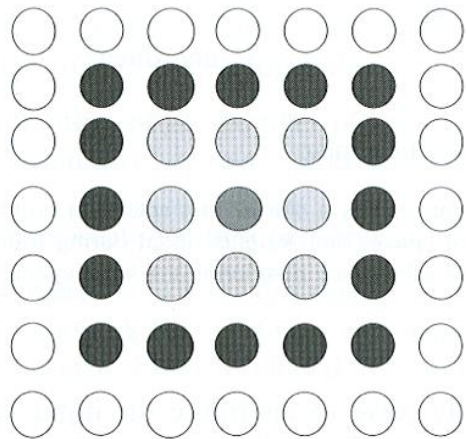
5. Decrease the learning rate and the size of the neighbourhood.
6. Go to 2.



Implementing the winning neighbourhood

Can choose equal weight updates for units within a certain distance from the winner or can multiply weight update with a neighbourhood function Λ that falls off with increasing distance (motivated by biological Mexican hat connectivity).

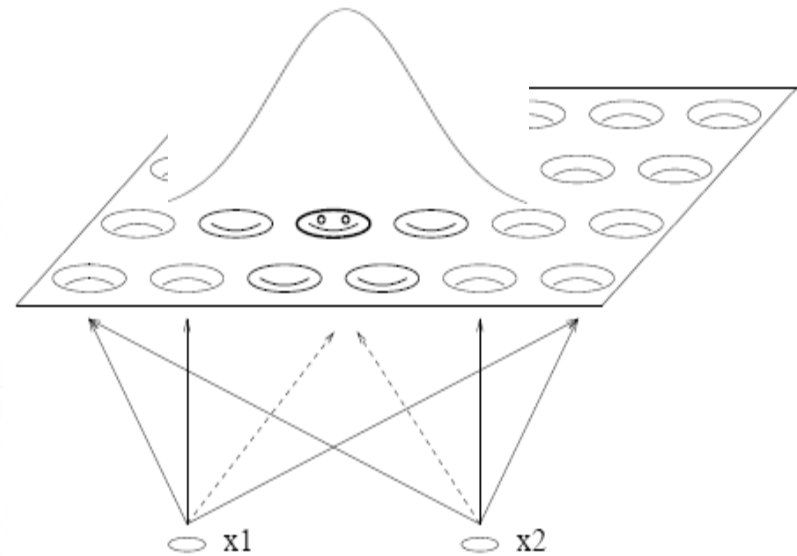
$$\vec{w}_i \rightarrow \vec{w}_i + \alpha \Lambda(i, i^*) (\vec{x} - \vec{w}_i)$$



 winning unit – only one to update if radius = 0

  units to be updated when radius = 1

   units to be updated when radius = 2



1D → 2D Kohonen example: the auditory cortex of a bat

Connectionism in Perspective,
(North-Holland, Amsterdam, 1989)
pp.403-412

Kohonen's Self-organizing Map for Modeling the Formation of the Auditory Cortex of a Bat

Thomas Martinetz, Helge Ritter and Klaus Schulten
Department of Physics
Technical University of Munich
D-8046 Garching

Abstract: As an example of Kohonen's self-organizing mapping algorithm we simulate the formation of a disproportionate representation of the ultrasound spectrum by the auditory cortex of a bat. We compare the results with experimental data of the cortex of the mustache bat *Pteronotus parnelli rubiginosus* and with an analytical calculation.

1. Determine lattice location $\mathbf{s}$ for which

$$||\mathbf{w}_{\mathbf{s}} - \mathbf{f}|| = \min_{\mathbf{r} \in A} ||\mathbf{w}_{\mathbf{r}} - \mathbf{f}|| \quad (2)$$

where $\mathbf{f}$ is the randomly chosen input signal for the current step.

2. For all sites $\mathbf{r}$ in the neighborhood of $\mathbf{s}$ (with $\mathbf{s}$ included) adjust

$$\mathbf{w}_{\mathbf{r}}^{(new)} = \mathbf{w}_{\mathbf{r}}^{(old)} + \epsilon h_{\mathbf{rs}}(\mathbf{f} - \mathbf{w}_{\mathbf{r}}^{(old)}), \quad (3)$$

where $0 \leq h_{\mathbf{rs}} \leq 1$ is a prespecified adjustment function of the distance $||\mathbf{r} - \mathbf{s}||$ and ϵ is a learning step size. $h_{\mathbf{rs}}$ has its maximum at $\mathbf{r} = \mathbf{s}$ and usually decays to zero, as $||\mathbf{r} - \mathbf{s}||$ increases.

Modelling the auditory cortex of a bat

Output space: a rectangle of 5 x 25 units that represent bat auditory cortex.

Input: a single ultrasound frequency that is draw from a probability distribution with a peak at 61 kHz (used by mustached bats for echolocation).

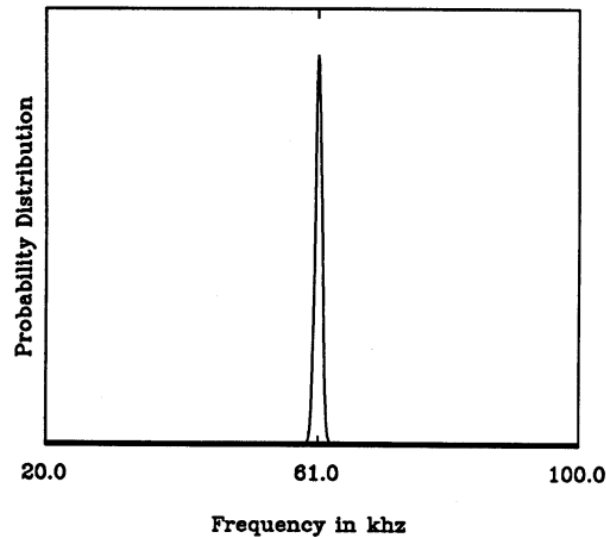


Fig.2 The probability distribution of the input signals versus the frequency of the signal, as assumed in the model. A Doppler shifted echo occurs three times as frequently as a signal of external ultrasound sources.

Formation of a 1D $\rightarrow$ 2D map in bat auditory cortex

Learning the map (output space):

78	45	63	43	75
51	77	28	86	91
52	80	24	34	45
52	85	91	85	84
23	97	26	27	37
96	28	98	86	82
44	99	50	81	42
52	86	74	92	75
66	39	56	41	34
99	23	77	69	61
42	47	81	38	100
93	86	56	45	23
36	70	46	59	22
62	59	80	97	25
64	92	84	23	84
83	59	31	89	47
76	71	41	41	89
72	27	85	27	67
51	23	93	100	27
89	52	32	42	65
48	44	30	65	45
21	34	83	77	31
25	38	82	38	79
47	75	43	86	64
20	73	43	51	64

Fig.3a The initial state of the neurons with their randomly chosen best frequencies. Each neuron is represented by a box displaying the associated best frequency. The neurons are arranged as a 5×25 -lattice.

28	28	30	32	35
28	29	31	33	36
30	31	32	35	37
33	33	35	37	39
36	37	38	39	41
40	41	41	43	44
44	44	45	46	47
48	48	49	49	50
51	52	52	52	53
54	54	55	55	55
57	57	57	57	57
58	58	58	58	58
59	59	59	59	59
60	60	60	60	60
60	60	60	60	60
60	60	60	60	60
60	60	60	60	60
60	60	60	60	60
61	61	61	61	61
62	61	61	61	61
62	62	62	62	62
63	63	63	63	62
64	64	63	63	63
64	64	64	64	63
65	64	64	64	64

Fig.3b The situation after 500 learning steps. At this stage the initial random correspondence between neurons and best frequencies has already given way to a tonotopic representation of the relevant frequency range.

36	33	29	26	25
38	35	32	30	27
40	39	37	35	33
44	42	41	39	39
47	46	44	44	44
50	49	48	48	49
54	53	53	54	54
58	58	58	58	58
59	59	59	59	59
60	60	60	60	60
60	60	60	60	60
61	61	61	61	61
61	61	61	61	61
61	61	61	61	61
61	61	61	61	61
62	62	62	62	62
62	62	62	62	63
62	62	63	63	63
64	64	64	64	65
68	68	68	69	68
72	73	73	74	73
75	77	79	81	81
78	80	84	87	88
81	83	87	91	93
82	85	90	94	96

Fig.3c The final state of the "auditive cortex" after 5000 learning steps. As a result of the narrow peak of the probability distribution of the sensory signals at 61kHz, a very disproportionate representation of the Doppler shifted echolocation signals has emerged. The associated neurons constitute almost half of the whole cortex.

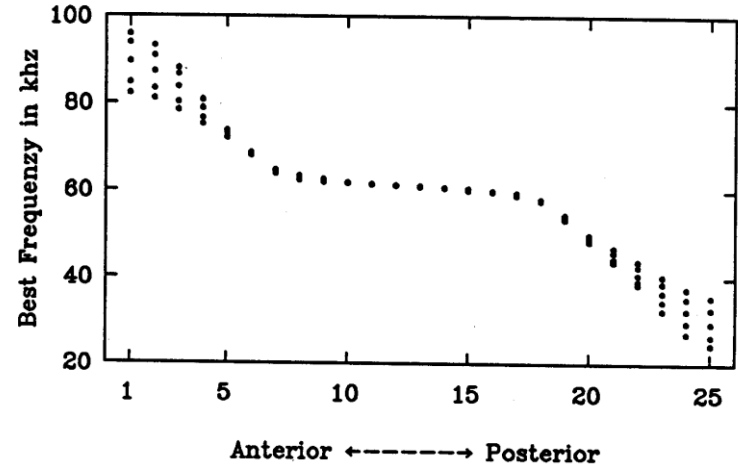


Fig. 4 A representation of the results of our computer simulation analogous to the representation of the experimental data in fig. 1C. The best frequencies of all neurons are plotted versus their coordinates 1 to 25 along the anteroposterior axis of our cortex model. For each position there are five neurons, forming a row in Fig. 3.

In the resulting tonotopic map, frequencies around 61 Hz are over-represented.

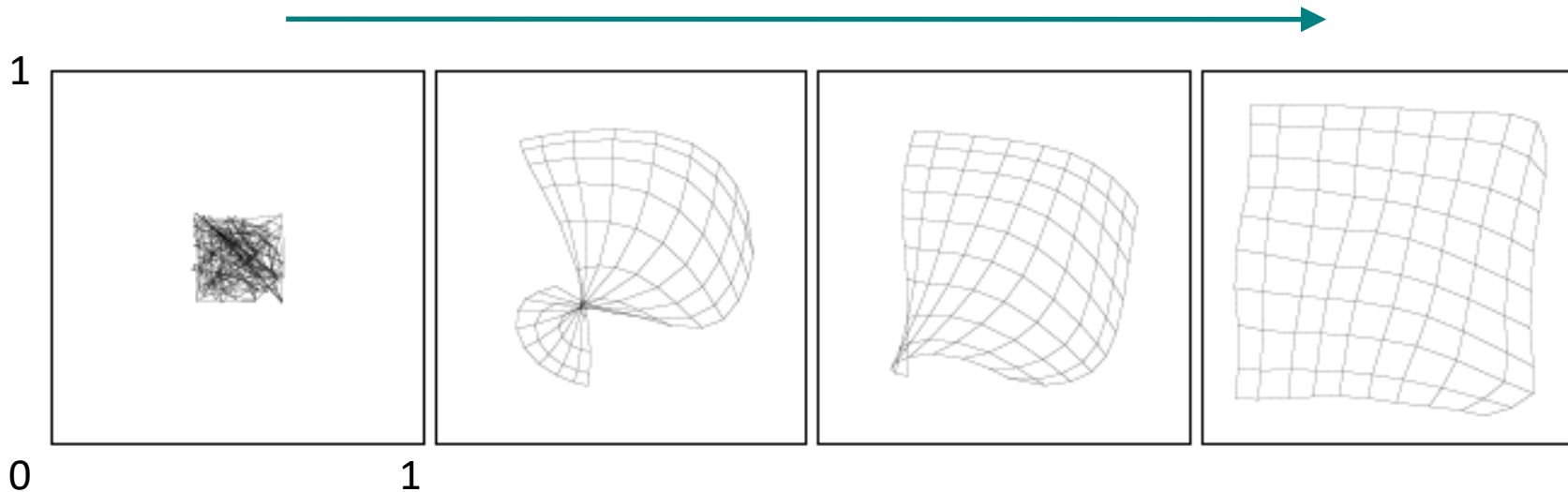
Formation of a 2D $\rightarrow$ 2D map with random input vectors

Output space: a 2D grid of output units.

Input space: input vectors (x_1, x_2) that are drawn randomly from a square region of a plane ($0 < x_1 < 1, 0 < x_2 < 1$).

The weight vectors start off at random locations close to $(0.5, 0.5)$.

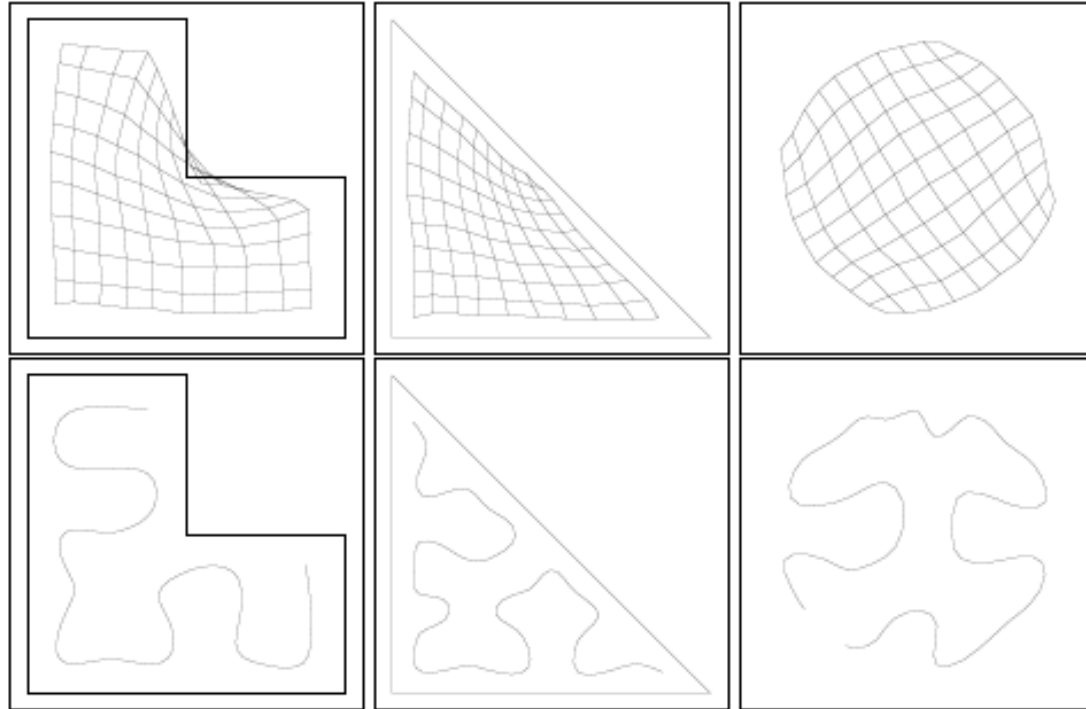
Learning pulls the weight vectors apart so that they cover the whole input space:



How could the final configuration be displayed in output space?

Other maps with random input vectors

2D $\rightarrow$ 2D maps



2D $\rightarrow$ 1D maps
(these result in
space-filling
Peano curves)

[demo](#)

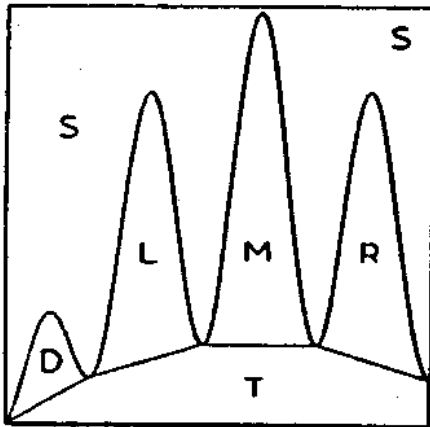
Modelling the formation of a somatosensory map

Output space: grid of 30 x 30 units that represent somatosensory cortex.

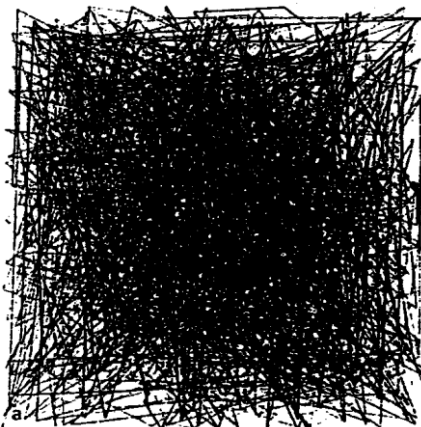
Inputs: vectors drawn randomly from five regions (D, L, M, R, T) of a hand-shaped surface.

Start with random weight vectors and apply Kohonen's algorithm (Ritter & Schulten 1986).

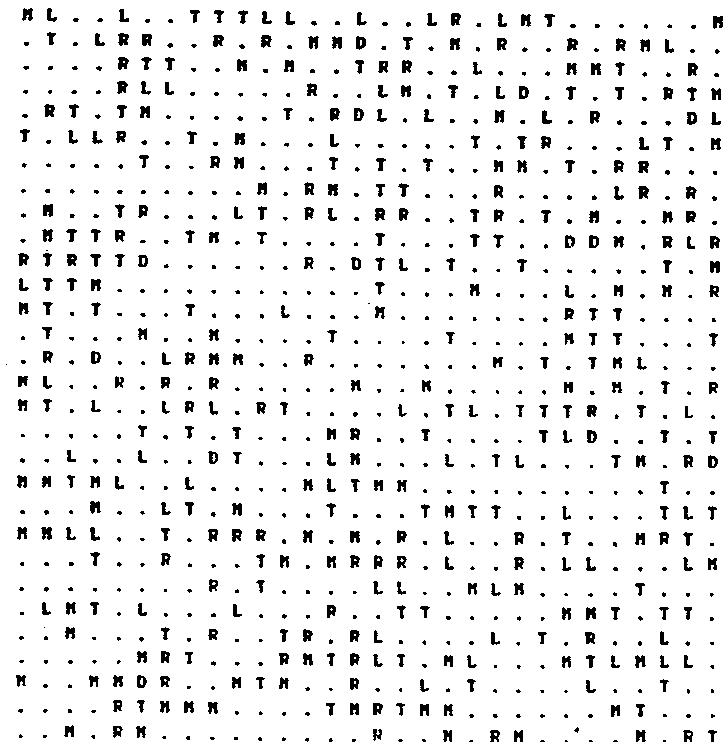
Input space



Initial
weight
vectors



Output space



Development of the somatosensory map

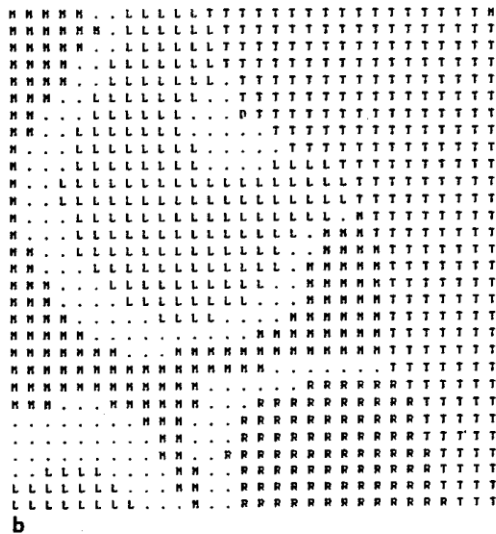
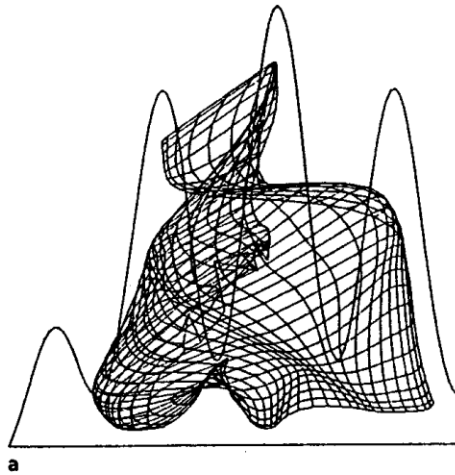


Fig. 4. a As Fig. 3a, but after 500 iterations. Superimposed is the hand area H, from which input signals v are originating. b As Fig. 3b, but after 500 iterations

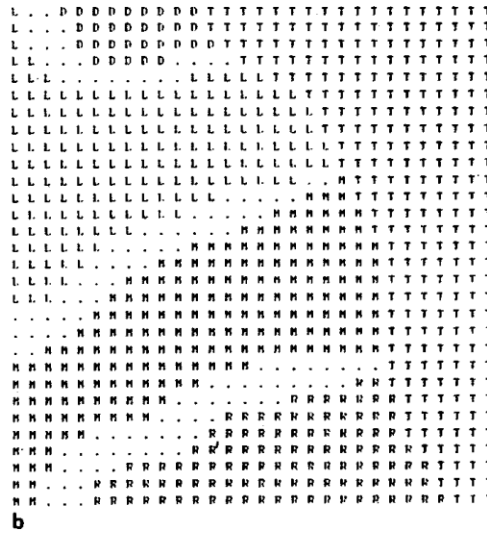
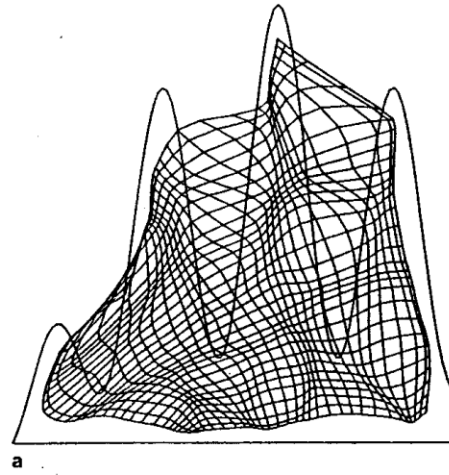


Fig. 5. a As Fig. 4a, but after 3000 iterations. b As Fig. 4b, but after 3000 iterations

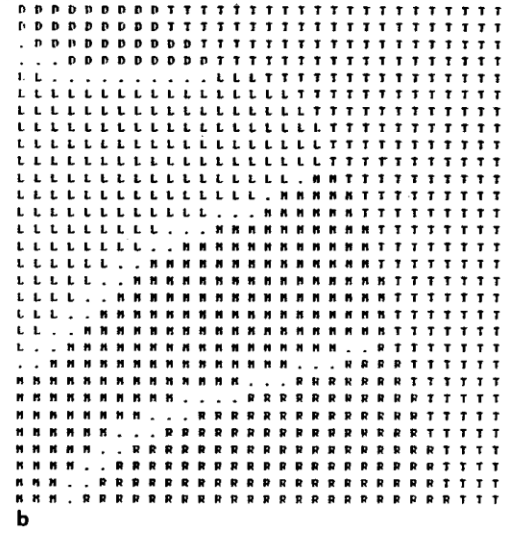
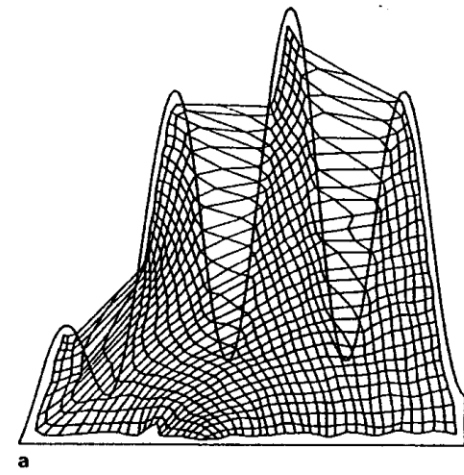
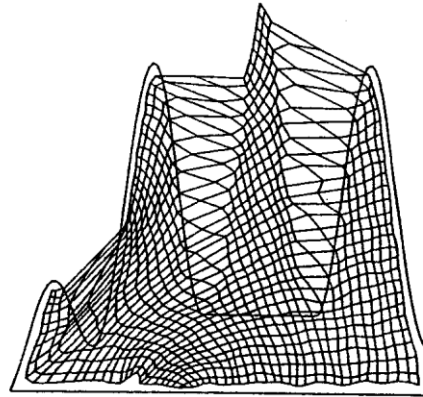


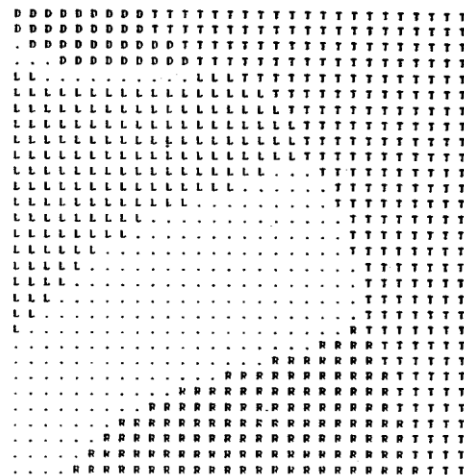
Fig. 6. a As Fig. 4a, but after 20000 iterations. b As Fig. 4b, but after 20000 iterations

Reorganization of the map in response to loss of a digit

Loss of input from one digit means that inputs from the neighbouring digits take over the inactive areas of simulated cortex.

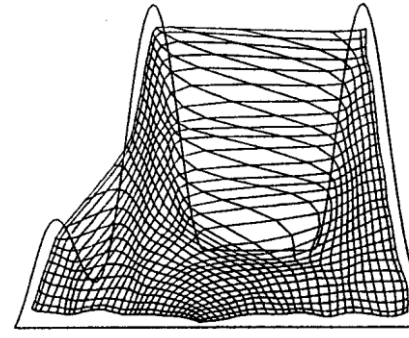


a

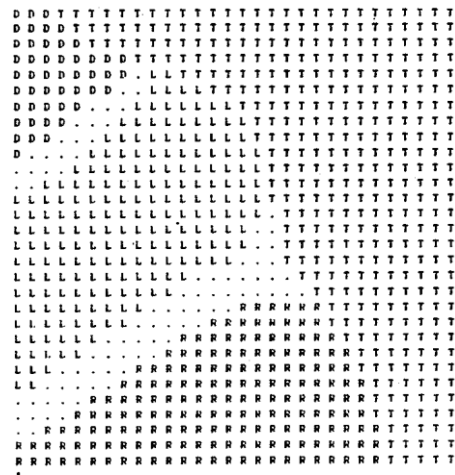


b

Fig. 7. a The same as the preceding figure, but now with region M excluded from contributing input signals v ; this situation corresponds to the removal of a finger. b Receptive field centers after removal of region M: units formerly responsive to finger M are now deprived of their inputs



a



b

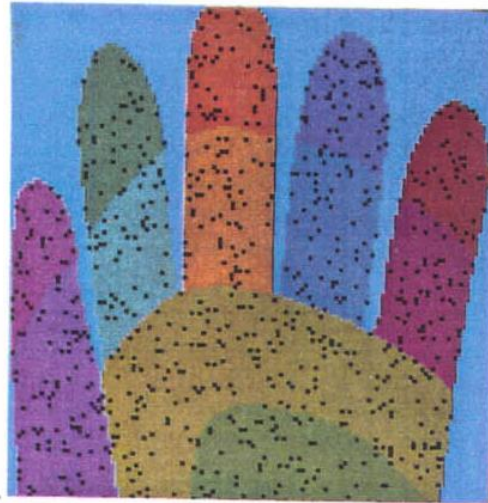
Fig. 8. a Readapted map after 50000 iterations subsequent to dissection of M. Formerly deprived units are re-employed by adjacent regions L, R, T thus allowing a finer representation there. b The same in "cortical view" as in Fig. 3b: formerly deprived units have developed receptive field centers located in neighbouring regions L, R, T

This replicates observations from experiments with monkeys (Merzenich 1983).

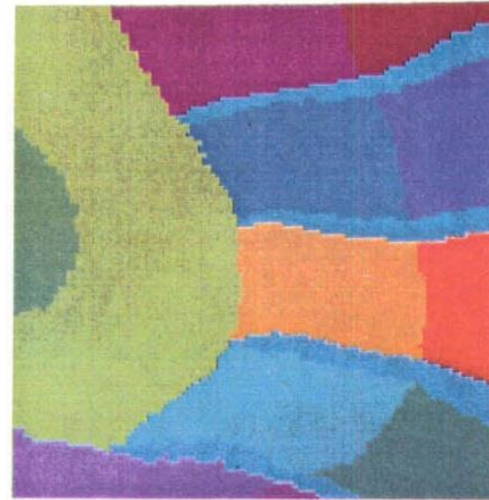
A large scale model of somatotopic map formation

800 receptors, 16,384 output units (Obermayer et al. 1990).

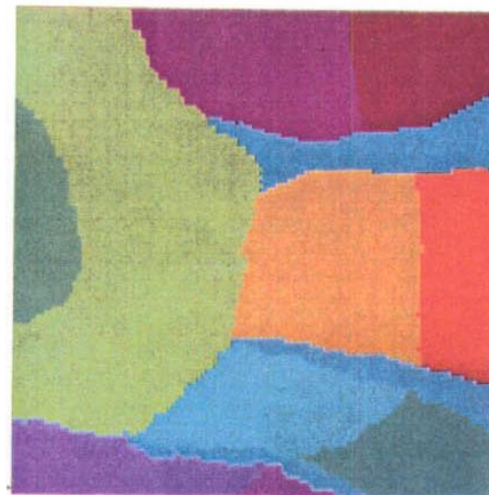
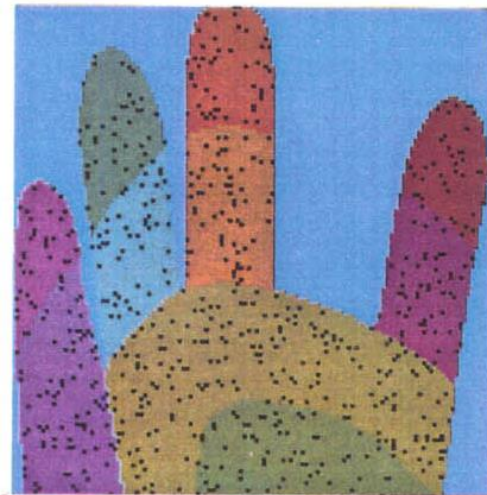
Input space



Final output space



Removal
of a digit



A self-organising semantic map

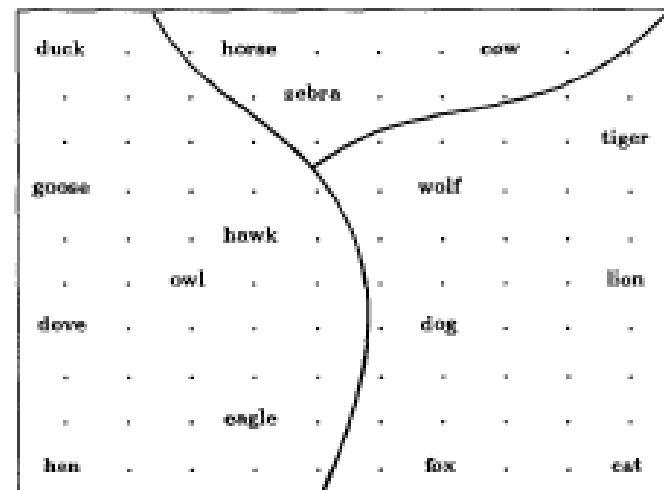
Ritter & Kohonen (1986): Output space: 2D grid with 10 x 10 output units.

Input space: binary vectors that encode features of animals.

	has-hair?	has-scales?	has-feathers?	flies?	lives in water?	lays eggs?
Dog	1	0	0	0	0	0
Cat	1	0	0	0	0	0
Bat	1	0	0	1	0	0
Whale	1	0	0	0	1	0
Canary	0	0	1	1	0	1
Robin	0	0	1	1	0	1
Ostrich	0	0	1	1	0	1
Snake	0	1	0	0	0	1
Lizard	0	1	0	0	0	1
Alligator	0	1	0	0	1	1

	dove	hen	duck	goose	owl	eagle	fox	dog	wolf	cat	tiger	lion	horse	zebra	cow
small	1	1	1	1	1	1	0	0	0	0	1	0	0	0	0
is medium	0	0	0	0	0	0	1	1	1	1	0	0	0	0	0
big	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1
2 legs	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0
4 legs	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1
has hair	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1
hooves	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1
mane	0	0	0	0	0	0	0	0	0	1	0	0	1	1	0
feathers	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0
hunt	0	0	0	0	1	1	1	0	1	1	1	1	0	0	0
likes run	0	0	0	0	0	0	0	0	1	1	0	1	1	1	0
to fly	1	0	0	1	1	1	1	0	0	0	0	0	0	0	0
swim	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0

Final output space:



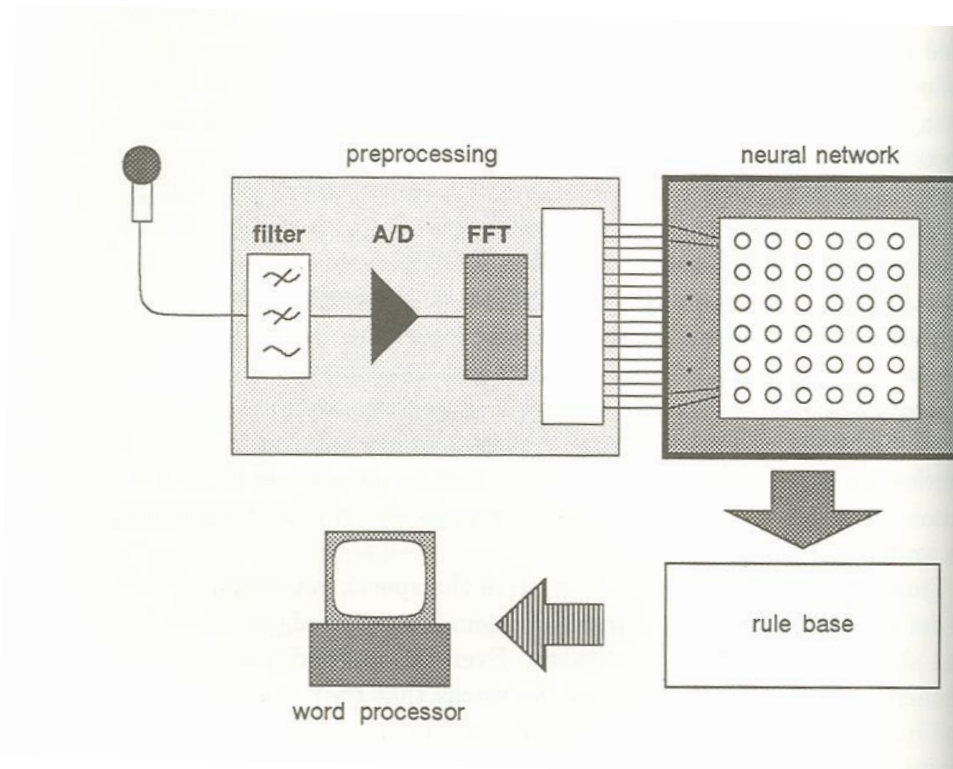
The resulting semantic map shows similarities between animal species (dots indicate weaker responses)

The phonetic typewriter

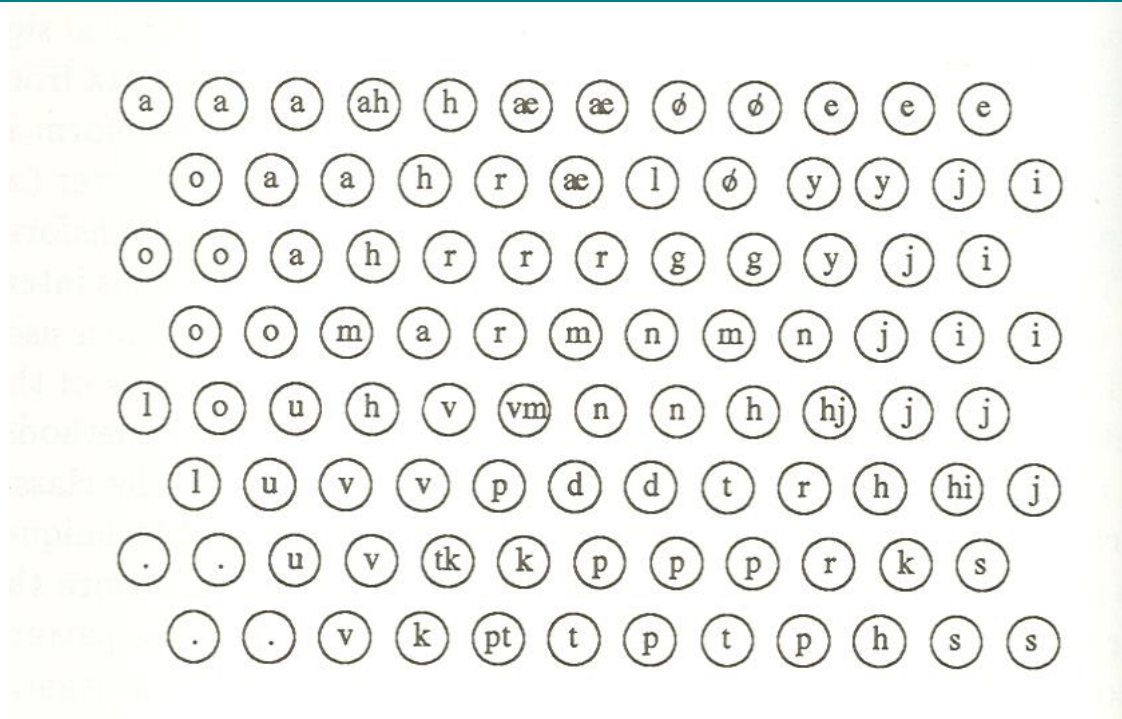
Developed by Kohonen as an automatic (Finnish) speech recognition system.

Preprocessing: divide speech waveform into short segments (9.83ms), perform FFT (Fast Fourier Transform) to generate 15-dimensional input vectors.

Use Kohonen algorithm to train weight vectors of a 2D grid of output units.



The phonotopic map of the phonetic typewriter



Learning results in output units that represents phonemes (sub-syllable components of speech), with similar phonemes close together in output space.

Problems: some units respond to more than one phoneme (for example /p, t/, /t, k/). Moreover, the pronunciation of phonemes is context-dependent and depends on the preceding and following phonemes (co-articulation effects).

Solution: use an auxiliary map to distinguish the phonemes /k, p, t/ and use a rule-based grammar processor to correct for co-articulation effects.

The travelling salesman problem

- What is the shortest tour around a given set of cities?
- No exact solution is known, cannot be solved by exhaustive search.

Number of cities

10

20

60

Number of possible tours

10^5

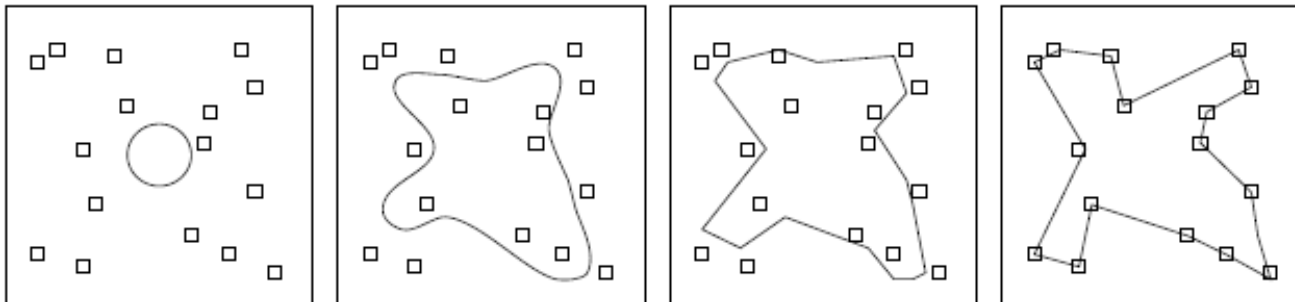
10^{30}

10^{80}

- Solving the travelling salesman problem is equivalent to finding a mapping from a 2D input space (position of the cities on the map) to a circular 1D output array (order of visiting the cities).
- Can be solved by a 1D Kohonen network or by the Elastic Net Algorithm (Durbin & Willshaw 1987).

The Elastic Net Algorithm (Durbin & Willshaw 1987)

- Maps a 2D input space (location of cities) to a 1D ring of output units (order of visit).
- Can use a simple physical analogy.
- Output units behave like beads on an elastic rubber ring.
- Beads are attracted by cities in the plane, with a force of attraction that decreases with distance.
- Elastic force between neighbouring beads on the rubber ring minimises the distances between successively visited cities.
- Maths is very similar to Kohonen algorithm.
- [Demo](#) (Yoshizawa 2001).



Comparison of the Elastic Net with Kohonen's learning rule

Kohonen's learning rule for weight vectors $\mathbf{w}_i$ and input vectors $\mathbf{x}$:

$$\vec{w}_i \rightarrow \vec{w}_i + \alpha \Lambda(i, i^*) (\vec{x} - \vec{w}_i)$$

The Elastic Net learning rule contains three terms: one for the attraction between units (beads) on the ring $\mathbf{w}_i$ and the position of the cities $\mathbf{x}_k$, and two others for the attraction between neighbouring units ($\mathbf{w}_{i-1}$, $\mathbf{w}_i$, $\mathbf{w}_{i+1}$):

$$\vec{w}_i \rightarrow \vec{w}_i + \alpha \left(\sum_k \Lambda_{ik} (\vec{x}_k - \vec{w}_i) + \beta (\vec{w}_{i+1} - \vec{w}_i) + \beta (\vec{w}_{i-1} - \vec{w}_i) \right)$$

Λ is a neighbourhood function and α and β are learning rates.

References

- Hertz, Krogh & Palmer. Introduction to the theory of neural computation. Addison-Wesley, 1991 (Chapter 9).
- Gurney. An introduction to neural networks. CRC press, 1997 (Chapter 8).
- Ritter, Martinetz & Schulten. Neural Computation and Self-Organizing Maps. Addison-Wesley, 1992 (Chapters 5-7).
- Durbin & Willshaw (1987). An analogue approach to the travelling salesman problem using an elastic net method. Nature 326, 689-91.

Exercises

- (1) If an input $\mathbf{x} = (1, 4)$ is presented to a network with two output units with weight vectors $\mathbf{w}_1 = (2, 3)$ and $\mathbf{w}_2 = (3, 2)$, which output unit will be the winner and why? Draw the vectors in input space and calculate the inner product of $\mathbf{x}$ with $\mathbf{w}_1$ and $\mathbf{w}_2$.
- (2) Competitive learning moves weight vectors in input space so that the weight vectors of different output units move to the centres of clusters in input space. Why is it beneficial for the weight vectors to move to the centres of the clusters?
- (3) Consider a Kohonen network with a 2D array of 5 x 5 output units and two inputs x_1 and x_2 that are chosen from a flat random distribution between 2 and 8. Draw the final state of the network both in input and output space.
- (4) Most self-organising maps map an N-dimensional input space onto a 2-dimensional (physical) output space. Give examples of 2D – 2D, 1D – 2D and 2D – 1D mappings.